GRADE 12 CHEMISTRY

EUEE MASTER STUDY GUIDE

Ethiopian University Entrance Exam
Complete Exam Preparation

Acid-Base Equilibria · Electrochemistry · Industrial Chemistry

Simplified · Exam-Focused · High-Score Strategies

UNIT 1

ACID-BASE EQUILIBRIA

Acids and bases are central to chemistry. This unit covers three definitions of acids and bases, equilibrium expressions, pH calculations, buffers, and titrations.

1.1 Acid-Base Concepts

Arrhenius Concept

Definition — Arrhenius Acid

A substance that produces H^+ ions (protons) in aqueous solution. Example: $HCl \rightarrow H^+ + Cl^-$

Definition — Arrhenius Base

A substance that produces OH^- ions in aqueous solution. Example: $NaOH \rightarrow Na^+ + OH^-$

Limitation: Only applies to aqueous solutions. Cannot explain bases like NH_3.

Brønsted-Lowry Concept

Definition — Brønsted-Lowry Acid

A proton (H^+) donor. Any substance that can give away H^+.

Definition — Brønsted-Lowry Base

A proton (H^+) acceptor. Any substance that can accept H^+.

Conjugate acid-base pairs:

When an acid donates H^+, the species remaining is its conjugate base. Example:

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

HCl is the acid; Cl^- is its conjugate base. H_2O is the base; H_3O^+ is its conjugate acid.

- The conjugate base has one fewer H and one more negative charge than the acid
- **Amphoteric substances:** can act as both acid and base (e.g. H_2O, HCO_3^-, HSO_4^-)

Lewis Concept

Definition — Lewis Acid

An electron pair acceptor. Does NOT need H^+. Example: BF_3, $AlCl_3$, Fe^{3+}

Definition — Lewis Base

An electron pair donor. Example: NH_3, H_2O, F^-

Advantage: Broadest definition — explains reactions without H^+ involved.

Arrhenius	Produces H^+ (aq)	Produces OH^- (aq)	Only for aqueous solutions
Brønsted-Lowry	H^+ donor	H^+ acceptor	Requires proton transfer
Lewis	e^- pair acceptor	e^- pair donor	Most general; no limitations

1.2 Ionic Equilibria of Weak Acids and Bases

Ionization of water:

$$H_2O \rightleftharpoons H^+ + OH^- \quad K_w = [H^+][OH^-] = 1.0 \times 10^{-14} \text{ at } 25°C$$

pH and pOH:

$$pH = -\log[H^+] \quad pOH = -\log[OH^-] \quad pH + pOH = 14$$

pH < 7 → acidic; pH = 7 → neutral; pH > 7 → basic/alkaline

Acid dissociation constant Ka:

$$\text{For weak acid } HA \rightleftharpoons H^+ + A^-: K_a = [H^+][A^-] / [HA]$$

Larger Ka = stronger acid. pKa = −log Ka

Base ionization constant Kb:

$$\text{For weak base } B + H_2O \rightleftharpoons BH^+ + OH^-: K_b = [BH^+][OH^-] / [B]$$

$K_a \times K_b = K_w = 1 \times 10^{-14}$ (for a conjugate acid-base pair)

Example 1.1:

Calculate the pH of 0.1 M HCl (strong acid, fully dissociates).

$[H^+] = 0.1$ M $= 10^{-1}$ M. pH $= -\log(10^{-1}) = 1$

Example 1.2:

Calculate the pH of 0.01 M NaOH (strong base, fully dissociates).

$[OH^-] = 0.01$ M. pOH $= -\log(0.01) = 2$. pH $= 14 - 2 = 12$

EXAM FOCUS

- pH calculations: always determine if acid/base is strong (fully dissociates) or weak (partial).
- pH + pOH = 14 always (at 25°C). This relationship is heavily tested.
- Know Ka for weak acids: larger Ka = stronger acid.

1.3 Common Ion Effect and Buffer Solutions

Definition — Common Ion Effect

The decrease in the ionization of a weak acid or base when a common ion is added from another source. Example: Adding NaCl to CH_3COOH — the Cl^- has no effect, but adding CH_3COONa suppresses ionization of acetic acid.

Definition — Buffer Solution

A solution that resists changes in pH when small amounts of acid or base are added. Contains a weak acid and its conjugate base (or weak base and conjugate acid).

Henderson-Hasselbalch equation:

$$pH = pKa + \log([A^-] / [HA])$$

Where $[A^-]$ = concentration of conjugate base, $[HA]$ = concentration of weak acid

How a buffer works:

- When H^+ added: $A^- + H^+ \rightarrow HA$ (conjugate base neutralises the acid)
- When OH^- added: $HA + OH^- \rightarrow A^- + H_2O$ (weak acid neutralises the base)
- Blood pH ≈ 7.4 maintained by carbonic acid / bicarbonate buffer (H_2CO_3/HCO_3^-)

Example 1.3:

What is the pH of a buffer made of 0.1 M CH_3COOH and 0.1 M CH_3COONa? $Ka = 1.8\times10^{-5}$

$pKa = -\log(1.8\times10^{-5}) = 4.74$. $pH = 4.74 + \log(0.1/0.1) = 4.74 + 0 = 4.74$

1.4 Hydrolysis of Salts

Definition — Hydrolysis

The reaction of an ion from a salt with water to form an acidic or basic solution.

Strong acid + Strong base	NaCl	Neutral (pH=7)	Neither ion hydrolyses
Weak acid + Strong base	CH_3COONa	Basic (pH>7)	CH_3COO^- hydrolyses → OH^-
Strong acid + Weak base	NH_4Cl	Acidic (pH<7)	NH_4^+ hydrolyses → H^+
Weak acid + Weak base	CH_3COONH_4	Depends on Ka, Kb	Both ions hydrolyse

1.5 Acid-Base Indicators and Titrations

Definition — Indicator

A weak acid (or base) that changes colour depending on the pH of the solution. The colour change occurs over a pH range of approximately 2 units.

Litmus	Red	Blue	6–8
Phenolphthalein	Colourless	Pink/Red	8.3–10
Methyl orange	Red	Yellow	3.1–4.4
Universal indicator	Red	Purple	1–14

Choosing the right indicator:

- Strong acid + Strong base: any indicator works (sharp endpoint)
- Strong acid + Weak base: use methyl orange (endpoint in acidic range)
- Weak acid + Strong base: use phenolphthalein (endpoint in basic range)

Titration calculation:

$n_1 = C_1V_1/1000$ moles of acid = moles of base at equivalence point

Example 1.4:

25 cm^3 of NaOH exactly neutralises 20 cm^3 of 0.5 M HCl. Find [NaOH].

Moles HCl = (0.5 × 20)/1000 = 0.01 mol. Moles NaOH = 0.01 mol.

[NaOH] = 0.01 × 1000/25 = 0.4 M

PRACTICE QUESTIONS

1. Write the conjugate base of: (a) H_2SO_4 (b) HCO_3^- (c) NH_4^+
2. Calculate the pH of: (a) 0.001 M HCl (b) 0.1 M NaOH (c) a buffer with pKa=4.74 and equal concentrations of HA and A^-
3. What is the pH of a solution of NH_4Cl? Explain using hydrolysis.
4. Choose the best indicator for: (a) HCl + NaOH (b) CH_3COOH + NaOH
5. 25 mL of H_2SO_4 neutralises 40 mL of 0.25 M NaOH. Find [H_2SO_4].

UNIT 2

ELECTROCHEMISTRY

Electrochemistry links chemistry and electricity. It covers oxidation-reduction reactions, the electrolytic cell (uses electricity to drive reactions), and the voltaic cell (produces electricity from reactions).

2.1 Oxidation-Reduction Reactions

Definition — Oxidation

Loss of electrons (or increase in oxidation number). OIL: Oxidation Is Loss

Definition — Reduction

Gain of electrons (or decrease in oxidation number). RIG: Reduction Is Gain

Memory aid: OIL RIG — Oxidation Is Loss, Reduction Is Gain

Rules for assigning oxidation numbers:

- Pure element: oxidation number = 0 (O_2, Fe, Na)
- Monatomic ions: oxidation number = ion charge (Na^+ = +1, Cl^- = −1)
- Oxygen = −2 (except in peroxides where it is −1)
- Hydrogen = +1 (with non-metals) or −1 (with metals/hydrides)
- Sum of oxidation numbers in a compound = 0

Balancing Redox Reactions — Half-Reaction Method:

1. Write the two half-reactions (oxidation and reduction)
2. Balance atoms (H and O using H^+ and H_2O in acidic solution)
3. Balance charges by adding electrons
4. Multiply half-reactions so electrons cancel
5. Add the half-reactions together

Oxidising and reducing agents:

- **Oxidising agent:** accepts electrons; is itself reduced
- **Reducing agent:** donates electrons; is itself oxidised

2.2 Electrolysis of Aqueous Solutions

Definition — Electrolysis

Using electrical energy to drive a non-spontaneous chemical reaction. Done in an electrolytic cell: two electrodes in an electrolyte connected to a power source.

Electrode reactions:

- **Cathode (−):** REDUCTION occurs here (positive ions gain electrons)
- **Anode (+):** OXIDATION occurs here (negative ions lose electrons)

Memory aid: CAtHODE = Reduction | ANOde = Oxidation → CAT walks in, AN ox walks out

Preferential discharge (selectivity):

When a solution contains multiple ions, which ones discharge first?

- **At cathode (cation reduction order):** $Cu^{2+} > H^+ > Pb^{2+} > Sn^{2+} > Fe^{2+} > Zn^{2+} > Al^{3+} > Na^+ > K^+$
- **At anode (anion oxidation order):** SO_4^{2-} and NO_3^- resist discharge; $OH^- > Cl^- > Br^- > I^-$ discharge preferentially

Electrolysis of dilute H_2SO_4:

- Cathode: $2H^+ + 2e^- \rightarrow H_2(g)$ [hydrogen gas produced]
- Anode: $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ [oxygen gas produced]
- Volume ratio $H_2 : O_2 = 2 : 1$

Electrolysis of brine (NaCl solution):

- Cathode: $2H^+ + 2e^- \rightarrow H_2(g)$
- Anode: $2Cl^- - 2e^- \rightarrow Cl_2(g)$
- Solution becomes NaOH
- Industrial uses: chlorine (disinfectant, PVC), hydrogen (fuel), NaOH (soap, paper)

Electrolysis of $CuSO_4$ solution (with copper electrodes):

- Cathode: $Cu^{2+} + 2e^- \rightarrow Cu(s)$ [copper deposits on cathode]
- Anode: $Cu(s) - 2e^- \rightarrow Cu^{2+}(aq)$ [anode dissolves]
- Used in electroplating and copper refining

2.3 Faraday's Laws of Electrolysis

Definition — Faraday's First Law

The mass of substance deposited or dissolved at an electrode is directly proportional to the quantity of electricity (charge) passed. $m \propto Q$

Definition — Faraday's Second Law

When the same quantity of electricity passes through different electrolytes, the masses deposited are proportional to their equivalent masses (molar mass/n).

Key formula:

$m = (Q \times M) / (n \times F)$ OR $m = (I \times t \times M) / (n \times F)$

m = mass (g), Q = charge (C) = $I \times t$, M = molar mass (g/mol)

n = electrons transferred per ion, F = Faraday constant = 96 500 C/mol

Example 2.1:

How many grams of Cu are deposited when 2 A flows for 30 min through $CuSO_4$?

$Q = I \times t = 2 \times (30\times60) = 3600$ C. $Cu^{2+} + 2e^- \rightarrow Cu$, so n=2.

$m = (3600 \times 63.5) / (2 \times 96500) = 228600 / 193000 \approx 1.18$ g

2.4 Industrial Applications of Electrolysis

- **Electroplating:** coating one metal with another to prevent corrosion or for decoration (chrome plating, silver plating, gold plating)
- **Electrorefining:** purifying impure metals (copper purification: impure Cu anode dissolves, pure Cu deposits on cathode)
- **Chlor-alkali process:** electrolysis of brine to produce Cl_2, H_2, and NaOH
- **Aluminium extraction:** electrolysis of molten Al_2O_3 (bauxite) in Hall-Héroult process
- **Anodising:** increasing thickness of natural oxide layer on aluminium for corrosion resistance

2.5 Voltaic (Galvanic) Cells

Definition — Voltaic Cell

An electrochemical cell that converts chemical energy into electrical energy using a spontaneous redox reaction. Two half-cells connected by a salt bridge.

Standard Electrode Potential (E°):

- Measured against the Standard Hydrogen Electrode (SHE), which is set to 0.00 V
- More positive E° → better oxidising agent → more likely to be reduced
- Cell voltage: $E°_{cell} = E°_{cathode} - E°_{anode}$
- Positive $E°_{cell}$ → spontaneous reaction → voltaic cell

Daniell Cell (classic example):

- Zinc electrode in $ZnSO_4$ (anode) + Copper electrode in $CuSO_4$ (cathode)
- Anode: $Zn \rightarrow Zn^{2+} + 2e^-$ (oxidation)
- Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$ (reduction)
- $E°_{cell} = E°_{Cu}\ (0.34\ V) - E°_{Zn}\ (-0.76\ V) = +1.10\ V$
- Salt bridge: maintains electrical neutrality by allowing ion flow

EXAM FOCUS

- Faraday calculations are frequently in EUEE — practise the formula m = ItM/nF.
- OIL RIG: Oxidation Is Loss, Reduction Is Gain. Cathode = reduction, Anode = oxidation.
- Know the difference: Electrolytic cell (uses electricity) vs Voltaic cell (makes electricity).

QUICK REVISION

- **Electrolytic:** cathode=reduction(−), anode=oxidation(+), needs power supply
- **Voltaic:** cathode=reduction(+), anode=oxidation(−), spontaneous, produces EMF
- **Faraday:** $m = ItM/nF$. $F = 96500$ C/mol
- **pH:** $pH=-\log[H^+]$; $pH+pOH=14$; $K_w=1\times10^{-14}$

PRACTICE QUESTIONS

1. What are the products at each electrode during the electrolysis of: (a) dilute H_2SO_4 (b) concentrated NaCl solution?
2. Calculate mass of silver deposited by 5 A for 1 hour (M Ag = 108 g/mol, n=1, F=96500 C/mol).
3. What is a salt bridge? What role does it play in a voltaic cell?
4. Assign oxidation numbers to each element in: $KMnO_4$, $K_2Cr_2O_7$, Na_2O_2.
5. Calculate E°cell for Zn/Cu cell. What makes it spontaneous?

UNIT 3

INDUSTRIAL CHEMISTRY

Industrial chemistry applies chemical reactions to produce useful substances on a large scale. Ethiopia has growing industries in many of these areas.

3.3 Manufacturing Key Chemicals

Ammonia (NH_3) — Haber Process

Equation: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ + heat (exothermic, reversible)

Industrial conditions:

- **Temperature:** 450°C — compromise (lower favours more NH_3 but reaction too slow; higher speeds up but yields fall)
- **Pressure:** 150–300 atm — high pressure favours NH_3 formation (4 moles gas → 2 moles)
- **Catalyst:** iron (Fe) with promoters Al_2O_3 and K_2O
- **Yield:** ~15% (unreacted gases recycled)
- **Uses:** fertilisers (urea, ammonium nitrate), nitric acid, explosives, cleaning agents

Sulphuric Acid (H_2SO_4) — Contact Process

Three stages:

1. Burn sulphur: $S + O_2 \rightarrow SO_2$
2. Oxidise SO_2: $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (V_2O_5 catalyst, 450°C)
3. Dissolve in H_2SO_4 → oleum, then dilute with water: $SO_3 + H_2O \rightarrow H_2SO_4$

- **Uses:** fertilisers, car batteries, paint, detergents, petroleum refining

Nitric Acid (HNO_3) — Ostwald Process

1. $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (Pt catalyst, 900°C)
2. $2NO + O_2 \rightarrow 2NO_2$
3. $3NO_2 + H_2O \rightarrow 2HNO_3 + NO$

- **Uses:** explosives (TNT, dynamite), fertilisers, dyes

Nitrogen-Based Fertilisers

- **Urea:** $CO(NH_2)_2$ — highest N content (~46%), widely used in Ethiopia
- **Ammonium nitrate (NH_4NO_3):** ~35% N; also explosive risk
- **Ammonium sulphate:** useful in acidic soils, adds S

3.4 Ethiopian Manufacturing Industries

Glass	Silica (SiO_2) + Na_2CO_3 + $CaCO_3$ melted at 1500°C	Packaging, construction, optics
Cement	Limestone ($CaCO_3$) + clay → clinker; add gypsum	Construction boom in Ethiopia; many factories (Mugher, Dangote)
Sugar	Sugarcane crushed → juice → evaporation → crystallisation	Fincha, Wonji-Shoa, Metahara factories
Ethanol	Fermentation: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (yeast)	Used as biofuel; Ethiopian Airlines uses some bioethanol blends
Soap/Detergent	Saponification: fat + NaOH → soap + glycerol	Traditional soap making widespread
Paper & Pulp	Wood cellulose dissolved, pressed, dried	Wonji paper mill
Tannery	Animal hides treated with tanning agents (chromium salts)	Ethiopia is major leather exporter

EXAM FOCUS

- Haber Process conditions: Fe catalyst, 450°C, 150-300 atm — all three are tested.
- Contact Process: V_2O_5 catalyst, 450°C — produces $SO_3 \rightarrow H_2SO_4$.
- Saponification equation and soap structure frequently appear in EUEE Chemistry.
- Know which Ethiopian industries produce which products.

PRACTICE QUESTIONS

1. Write the equation for the Haber process. State the conditions used and explain why each is chosen.
2. Describe the three stages of the Contact Process for making H_2SO_4.
3. Write the equation for saponification. What products are formed?
4. What is the role of the catalyst in the Ostwald Process? Name it.
5. Explain why ammonia yield in the Haber Process is relatively low despite optimal conditions.

LAST-MINUTE REVISION GUIDE

pH formula	**$pH = -\log[H^+]$; $pH + pOH = 14$; $K_w = 1\times10^{-14}$**
Strong acids	**HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_4$ — fully dissociate**
Weak acids	**CH_3COOH, HF, H_2CO_3 — partially dissociate, use Ka**
Henderson-Hasselbalch	**$pH = pKa + \log([A^-]/[HA])$**
Buffer blood	**H_2CO_3/HCO_3^-; maintains blood pH ≈ 7.4**
OIL RIG	**Oxidation Is Loss; Reduction Is Gain**
Electrolysis cathode	**Reduction (cations gain e^-)**
Electrolysis anode	**Oxidation (anions lose e^-)**
Faraday formula	**$m = ItM / (nF)$; $F = 96500$ C/mol**
Brine electrolysis	**Cathode: H_2; Anode: Cl_2; Solution: NaOH**
Voltaic cell	**$E°cell = E°cathode - E°anode$; positive = spontaneous**
Haber Process	**$N_2+3H_2 \rightleftharpoons 2NH_3$; Fe catalyst, 450°C, 150-300 atm**
Contact Process	**$SO_2+\frac{1}{2}O_2 \rightleftharpoons SO_3$; V_2O_5 catalyst, 450°C**
Saponification	**Fat + NaOH → Soap + Glycerol**
Salt hydrolysis	**Weak acid salt → basic; Weak base salt → acidic**

EUEE EXAM STRATEGY

Time Management

- **Skim first (5 min):** Read all questions. Mark easy (E), medium (M), hard (H).
- **Easy first:** Answer all E questions to bank safe marks and build confidence.
- **Pace yourself:** Aim for ~2 minutes per question. If stuck over 3 minutes, skip and return.
- **Show working:** Even wrong answers earn partial marks if steps are correct.
- **Final review:** Use last 10 minutes to check calculations, names, and dates.

Answering Techniques

- Multiple-choice: eliminate obviously wrong answers first, then choose from what remains.
- Definition questions: always include the key term in your answer.
- Explanation questions: structure as Definition → Example → Importance.
- Diagram questions: label all parts, use arrows correctly, add units where needed.
- Essay questions: write a clear intro, 3-4 main points with evidence, strong conclusion.

Common Exam Traps

- Confusing similar terms — always define the exact term asked.
- Incomplete answers — re-read every question before finalising.
- Time overrun — if a question takes too long, note it and move on.
- Skipping diagrams — a well-labelled diagram often earns more marks than a paragraph.
- Not reading all options in MCQ — always read all four choices before selecting.

NEXT LEVEL LEARNING

Quantum Chemistry & Molecular Orbital Theory

University chemistry explores electron behaviour using quantum mechanics. Molecular orbital theory describes bonds more accurately than simple covalent bonding — electrons exist in molecular orbitals spread across the whole molecule.

Advanced Thermodynamics

You will study Gibbs free energy ($\Delta G = \Delta H - T\Delta S$) which predicts whether a reaction is spontaneous. $\Delta G < 0 \rightarrow$ spontaneous; $\Delta G > 0 \rightarrow$ non-spontaneous; $\Delta G = 0 \rightarrow$ equilibrium.

Organic Chemistry

A full year of university chemistry focuses on organic reactions: nucleophilic substitution, elimination, addition, and aromatic chemistry. These form the basis of pharmaceuticals and materials science.

Biochemistry

Biochemistry bridges biology and chemistry — studying how enzymes, DNA, and proteins function at the molecular level. Enzyme kinetics (Michaelis-Menten) and metabolic pathways are key topics.